

Illicit Drug Substitution and Crime Dynamics: Evidence from the Taliban’s Narcotics Ban

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Abstract

Understanding how illicit drug markets and associated social harms respond to exogenous shocks is critical for effective policy formulation. The secretive nature of illicit drug markets makes their study especially difficult. The Taliban’s Afghanistan ban on opium production in April 2022 provides a natural experiment for examining drug substitution behaviors and their social consequences. By analyzing international opium price fluctuations resulting from this ban and employing wastewater-based epidemiology to track consumption changes in Australian communities, I identify shifts in drug use patterns. Furthermore, I explore the impact of these market disruptions on community crime rates. I explore how the effects of presumed increases in financial pressures on users from supply shocks manifest in community crime rates. The findings offer valuable insights into the adaptive behaviors of drug users amid market shocks and contribute to the evaluation of policy interventions aimed at mitigating harm in illicit drug markets.

Keywords: Illicit Markets, drugs, heroin, crime, supply shocks **JEL Codes:** K42, D12, E26, I18, F14

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1 Introduction

Illegal drug markets are large and highly consequential for society. Estimates from the United Nations Office on Drugs and Crime (UNODC) put the size of the global illegal market at between \$400-500 billion per year, representing an estimated 316 million users across all drug types ¹. Beyond the associated loss of government tax revenue from illegal market activities, these markets generate substantial negative externalities, including overdoses (Zoorob 2019), adverse long-run health outcomes (Nelson et al. 2011, Mathers et al. 2013,), and violence (Dell 2015, Castillo, Mejía, and Restrepo 2020, Fryer Jr. et al. 2013). What many policymakers describe in broad terms as “social problems” are, at their core, outcomes of a market in which firms produce and distribute products, incur and respond to costs, and collect revenue while consumers, who are budget constrained, seek to maximize utility from consumption.

Historically, governments have combated illegal drug markets primarily through punitive approaches. Policies targeting these markets have emphasized arresting dealers and users, disrupting trafficking organizations, and interdicting shipments, with the ultimate goal of reducing the supply and availability of targeted substances. The effectiveness of such interventions depends crucially on how responsive market participants are to changes in market conditions. When demand is inelastic, large supply shocks may raise prices and increase financial strain on users without meaningfully reducing consumption. Furthermore, with available substitutes, users may just switch consumption to another undesirable product as the price or availability of a certain product becomes too much of a strain. As outlined in Becker, Murphy, and Grossman 2006, such conditions can even increase profitability for surviving suppliers that evade enforcement, potentially strengthening remaining criminal organizations and intensifying the harms associated with prohibition.

In this paper, I estimate the consumption responses of drug users in Australia to a large, unexpected supply shock to international heroin markets. In April 2022, the Taliban in Afghanistan announced a ban on the production and consumption of narcotics within the country. Because Afghanistan has historically supplied the bulk of the global world’s opium, this this decree translated into an approximate 80% reduction in global opium supply.² This shock represents what is likely the largest single drug market intervention in modern history and was followed by a roughly ten-fold increase in the farm-gate price of opium. I use this increase in the international price of opium to measure the severity of the shock, allowing me to estimate the own-price elasticity of demand for heroin as well as cross-price elasticities with other drug types. I then link these estimated elasticities to local crime outcomes, exploiting the timing of the price movements to study effects of increased financial strain on on users of drugs to different categories of crime.

Generally, following the large increase in international prices, I find reduction in heroin consumption with lower reductions in consumption from areas with higher pre-ban levels of heroin consumption. I also document substitution into other drugs, most notably fentanyl, suggesting that some consumers respond by switching toward more potent and potentially more lethal substances. On the crime side I find increased

¹estimates from UNITED NATIONS OFFICE ON DRUGS AND LABOR 2025 and Reuter and V. Greenfield n.d.

²United Nations Office on Drugs and Crime 2022.

rates of petty theft and other economically motivated crime, but no detectable aggregate effect on violent or personally motivated crimes. These results underscore the role of financial pressure in shaping individuals' propensity to engage in acquisitive crime, and they highlight how supply-side interventions in drug markets can shift rather than resolve underlying harms, both in terms of substance use and criminal activity.

The Taliban's 2022 opium ban is an ideal natural experiment for studying illegal drug market responses to large shocks. It was substantial in magnitude, largely unanticipated, and has so far been persistent. In many respects, it mimics the idealized objective of aggressive law enforcement interdiction campaigns where a targeted drug is suddenly and sharply reduced from global supply. Observed market reactions to this ban can therefore be informative about the likely consequences of more conventional enforcement strategies aimed at reducing supply in illegal drug markets.

The findings from this paper contribute to three main strands of the literature: (i) price elasticities in illicit drug markets, (ii) market responses to exogenous supply shocks, and (iii) the link between drug markets and crime, especially economically motivated offending by users. A unifying challenge across all three is that researchers must infer behavior in illegal markets from imperfect proxies such as self-reports, seizures, health outcomes, or arrest data. Each of these possess their own biases and complicate our ability to fully assess which policies "work" and through what mechanism.

A first strand focuses on price elasticities in illegal drug markets. Understanding how drug consumption responds to price, and how users substitute across substances when relative prices change, is a key factor for predicting the effects of enforcement, interdiction and regulatory policies. Gallet 2014 provides a comprehensive meta-analysis of 42 studies reporting 462 elasticities, including some positive estimates, and shows that results are highly sensitive to methodological choices, samples, and outcome measures. His preferred meta-regression suggests a heroin price elasticity of roughly -0.51, but the broader takeaway is the substantial uncertainty around elasticity estimates in illegal markets, where prices, enforcement, and measurement error move together.

Much of the uncertainty arises from the proxies used to measure demand. For instance Saffer and Chaloupka 1999 and Grossman, Chaloupka, and Brown 1996 combine self-reported national survey data with DEA seizure-based price data to estimate relatively large price elasticities for heroin and cocaine (-0.94 and -1.10 respectively). However, their outcome variables are typically binary indicators of any use rather than determinants for the level of consumption in the population. This means the estimated elasticities reflect entry and exit from use, not changes at the intensive margin, and are potentially sensitive to misreporting influenced by social stigmas around drug consumption or fear of legal repercussion.

A second common strategy uses health outcomes such as overdoses or hospital admissions as proxies for aggregate consumption. Unick et al. 2014 document a negative relationship between the purity adjusted price of heroin and heroin-overdose hospitalizations, estimating that a \$100 decrease in the price of pure heroin increases overdose hospitalizations by 2.9%. Cunningham and Finlay 2016 exploit price increases driven by regulations on precursor chemicals for methamphetamine to estimate demand responses and find

relatively low elasticities around -0.24 to 0.28 for hospital admissions. These approaches typically rely on purity-adjusted-price (price divided by purity) as the relevant consumer price. Health harms associated with drug use are themselves a function of purity. As shown by Darke et al. 1999, overdose risk is closely linked to drug purity. When price, purity, and health outcomes co-move, it becomes difficult to disentangle demand shifts from biological responses to changes in potency. Together, this literature highlights not only wide variation in elasticity estimates, but also how reliance on noisy proxies can obscure the true behavioral response to policy and price changes.

A second literature examines how illicit drug markets respond to exogenous supply shocks, often generated by changes in regulation or enforcement of precursor chemicals. These quasi-experimental settings are attractive because they help to isolate consumer behavior from broader market trends. Cunningham and Finlay 2016 study U.S. laws tightening control over precursor chemicals for methamphetamine and track impacts across several proxies of use, including hospital admissions and self-reported treatment entries. They find modest declines in methamphetamine use, suggesting that even large input-side restrictions may generate smaller-than-expected behavioral responses. Dobkin and Nicosia 2009 exploit the same 1995 U.S. intervention in pseudoephedrine markets and find roughly a 55% drop in methamphetamine use among arrestees, with little evidence of a change in property or violent crimes associated with the intervention. These studies once again rely on proxies to estimate demand changes which make the interpretation of effect sizes dependent on how closely these measures track underlying consumption versus changes in detection, treatment access, or reporting.

More closely related to this paper are studies that trace consumer responses to large shocks in opioid markets. Moore and Schnepel 2021 analyze the Australian “heroin drought” of 2001 which was characterized by a sudden contraction of heroin availability. They find substantial short-run reductions in overdoses, implying a decline in heroin use, but these gains appear short lived as users substitute into other substances and crime increases. While similar in setting, a key difference is that the 2001 Australian episode involved a clear physical shortage of heroin within the country. These shortages were widely reported, but the 2022 shock does not reduce availability. Observed effects are therefore more plausible driven by financial mechanisms rather than outright scarcity.

Deiana, Giua, and Nisticò 2024 take a more global perspective, examining price shocks to Afghan opium markets in the early 2000s and linking them to changes in opioid prescribing in U.S. counties. They show that declines in Afghan opium prices are associated with higher levels of prescribed opioids in the U.S. underscoring how international supply conditions can ripple through legal and illegal segments of opioid markets.

A third strand of the literature studies the relationship between drug markets and crime. Paul J. Goldstein 1985 outlines how drug use can influence crime through three channels: (i) direct pharmacological effects on behavior, (ii) systemic violence within illegal distribution networks, and (iii) “economic-compulsive” crime when users commit offenses to finance consumption. Existing work has documented strong correlations

between drug involvement and property crime (), but pinning down causal mechanisms is often complicated by confounders such as local labor market conditions, policing intensity, and unobserved preferences.

By combining an exogenous international supply shock with granular measures of local consumption and crime, this paper adds to the evidence on the economic-compulsive channel. In particular, I examine how a plausibly exogenous increase in the effective price of illicit opioids affects the rate of property and other economically motivated crimes by users, in a context where physical availability of the drug does not appear to decline. This allows me to separate the scarcity-driven mechanism from the budgetary pressure one and show that price induced financial pressure on users can translate into measurable changes in crime. In doing so, the paper underscores both the centrality of economic motives in drug-related crime and the need to interpret policy evaluations cautiously when they are inferred from policy outcomes rather than direct measures of consumption and welfare.

Finally, I contribute to the small, but growing, literature that uses wastewater-based epidemiology (WBE) data to measure drug consumption. Wastewater monitoring provides an anonymous, population-level measure of consumption based on concentrations of drug metabolites in sewage, aggregated over catchment areas. This approach offers a direct, continuous proxy for aggregate consumption and helps overcome some of the identification issues associated with surveys, seizures, and health outcomes in illicit markets.

While wastewater data have been used extensively in public health, epidemiology, analytic chemistry and other related STEM fields, application in broader social science contexts are limited. Previous studies typically rely on detection of drugs rather than analyzing trends in flows. Daniele, Soliman, and Vargas [n.d.](#) is the first to use wastewater data to directly measure drug consumption. They study supply shocks in Colombian cocaine markets and link these shocks to violence along supply routes and identify consumption changes via wastewater monitoring in Europe. Using data from the European Union Drug agency (EUDA), they show that increase in cocaine production in Latin America are followed by increases in consumption across European countries, with larger effects in countries more tightly connected to Colombian supply chains. However, the EUDA's wastewater data are limited both in the range of drugs analyzed and in the frequency of testing compared to the Australian data used in this paper.

The Australian wastewater monitoring program has been widely used in various fields, but less frequently connected to economic behavior and market mechanisms. B. Tschärke et al. [2026](#) examine the effects of an alcohol sales ban on alcohol consumption in Alice Springs, Northern Territory, using wastewater data to capture changes in population-level drinking, highlighting the power of this data to capture first-stage effects. Wang et al. [2026](#) use the same data to study changes in tobacco consumption in the context of a growing illegal tobacco market in Australia, documenting a shift toward e-cigarette use that increasingly accounts for consumption in the illegal market. Verhagen et al. [2026](#) focus on geographic determinants of drug use, showing that remoteness and community characteristics significantly influence cannabis consumption rates. These papers highlight the strength of wastewater data in capturing changes in population level consumption, but only connect the findings to social outcomes in a limited way.

I am the first, to my knowledge, to use wastewater data to measure market-level impacts of a large international supply shock across a broad set of drugs. I am also the first to use direct, population-level, consumption measures to estimate price elasticities of drugs. Using this data I am also able to use pre-shock wastewater consumption measures to control from geographic variation in baseline intensity of drug use across localities. This allows me to show that markets may differ systematically in their responses to a common shock depending on preexisting consumption patterns and local characteristics. This is an idea well established in the broader economics literature but largely absent in empirical work on illicit drug markets. In this way, the paper not only speaks to debates about price elasticities, supply shocks, and crime, but also illustrates how improved measurement via WBE can be used to supplement or replace proxies and sharpen inference about which policies are effective.

The remainder of the paper is organized as follows. Section 2 provides background on the Taliban’s opium ban and its impact on global opiate markets. Section 3 describes the wastewater monitoring program and the construction of my consumption measure. Section 4 provides a simple theoretical framework linking drug price changes to consumption and crime via budget constraints and substitution. Section 5 describes the data on wastewater, drug prices, and crime. Section 6 details the empirical strategy used to estimate price elasticities and crime effects. Section 7 presents the main results, and section 8 concludes with a discussion of policy implications and directions for future research.

2 Background

2.1 Taliban Narcotics Ban

After nearly 20 years of conflict, the United States and Afghanistan signed a peace accord in 2020 officially signifying the United States intention to leave the country. The U.S. completed its evacuation on Aug, 30, 2021. Following the U.S.’s departure, the Taliban quickly regained control of the country, becoming the de-facto government.

In April of 2022, the Taliban announced a ban on consumption and cultivation of narcotics within the country C. Greenfield and Ahmad 2022. The implications of this ban would be tremendous on the opium crop produced within the country. In 2021, Afghanistan accounted for an estimated 86% of global illicit opium production. At the time of the announcement, many were skeptical the ban would hold. For starters, revenue from the illicit opium market represented 9-14% of the countries GDP in 2021 ³. Many poor farmers relied on revenue from the crops to supplement their own income as well so enacting the ban would put the country economically at risk as well as increasing hardships for some of its poorest citizens.

In addition to the economic and political conditions within the country, many cited the Taliban’s previous ban in 2000, which proved short-lived, as evidence that the ban could not hold. Nonetheless in 2023, the year of the next growing season following the ban’s announcement, cultivation of opium in Afghanistan was down 95%. The ban has remained in effect since its announcement with cultivation consistently lower compared

³Shehryar Fazli 2022

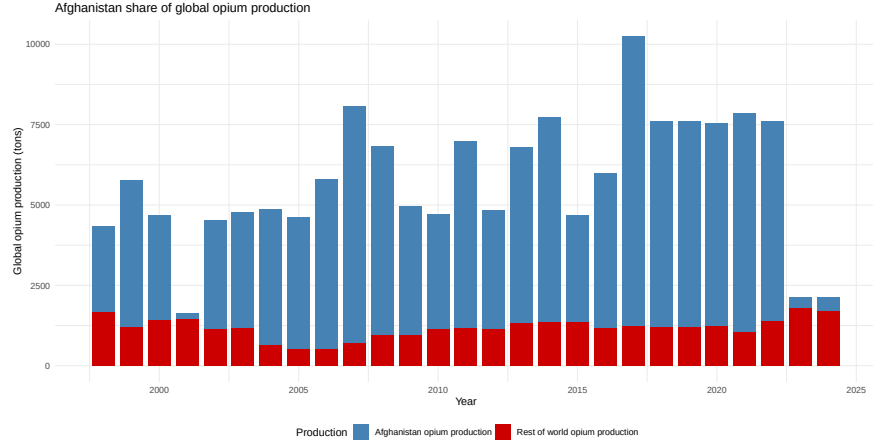
to pre-ban levels.

Figure 1 highlights the impact the ban had on global opiate markets. Panel (a) shows Afghanistan’s share of the global opium market from 1998-2024. Afghanistan has always been a vitally important country in the global opiate market accounting for the majority of production. Prior to the ban, Afghanistan produced was estimated to produce 6,800 tons of raw opium accounting for approximately 86% of the global total 7,850 tons. In 2023, the country produced an estimated 333 tons with global production at only 2,130 tons. While Afghanistan still has a presence in the illicit opium market, accounting for approximately 20% of global production, the steep decline in output of the world’s largest producer has lowered production by approximately 73%.⁴

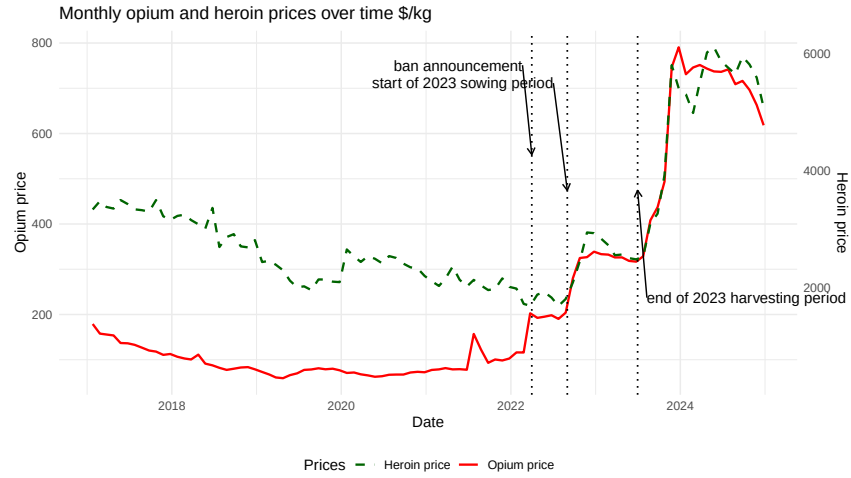
Panel (b) plots the monthly wholesale prices for farm-gate opium and heroin. Both are listed in USD per kilogram. Prior to the 2022 ban, prices of raw opium remained relatively stable and wholesale heroin prices even had small downward trends. Faced with the fear of supply shortage, farm-gate prices of dried opium rose from \$202/kg in March 2022, to a high of \$790/kg in December of 2023, a 291% increase. Wholesale prices for heroin similarly increased from \$1693/kg to \$5405/kg over the same time period, a 219% increase. The price increases accelerated following the 2023 growing season, possibly as traders realized the efficacy of the ban in Afghanistan. Heroin prices, while smaller in absolute percentage changes, closely follow the patterns of opium as it is the primary ingredient in the production of heroin. There is a slight pre-trend in the two months prior to the announcement possibly indicating some level of insider-awareness of the coming ban although no reporting of widespread awareness has been documented.

In response to the shortage, and subsequent price increase, other countries have increased production, presumably to capitalize on the increased margins. Myanmar, now the world’s largest illicit opium producer, increased domestic production by 35% from 2022 to 2023. Despite these increases the world’s supply is still at a steep shortage.

⁴The ban granted a small concession to farmers, giving farmers several months to harvest and sell their current year’s poppy crop. This resulted in the ban primarily impacting the following year’s sowing behavior. As such, the full extent of the ban’s impact did not become apparent to many until the following year’s harvesting season in spring of 2023.



(a) Global Opium Production



(b) Opium and Heroin prices \$/kg

Figure 1: Afghan Opium Ban

Notes: Panels show impacts of the Taliban's opium ban on output and prices. All production and price data is from UNODC. Panel (a) shows global opium production by year from 1998-2024. Afghanistan's share of global production the blue (upper) section of the bar with all other countries production in grey (below). Panel (b) plots monthly wholesale prices for opium and heroin within Afghanistan. All prices are in USD. From left to right, the dotted lines correspond to the announcement of the ban, the start of the 2023 poppy sowing season, and the end of the 2023 opium harvesting season respectively.

Despite the clear decrease in new production of opium for the world market, there are reasons to suspect retail markets would not be hit immediately by a shortage of the drug. Reports from the EUDA and UNODC document enough reserves from stockpiling to meet world demand in the short run. The increases in prices are then likely reflecting monetization of expected future prices. While some retail markets might see supply shortages as wholesale sellers seek to outbid each other for these stockpiles, changes in consumption are more likely to be driven the diffusing of increasing costs through the supply chain.

2.2 Drug Market in Australia

Australia’s drug market can be representative of Western Europe or the United States in many way, but it also has distinct features which may hinder extrapolation of effects onto other nations and regions. For one, Australia is geographically isolated. The country shares no land borders with other nations. As a result of it’s seclusion, all drugs produced outside the country need to enter through shipping or air cargo. A substantial amount of the drugs not produced within Australia are imported from the region known as the “golden triangle” which covers Thailand, Myanmar, and Laos.

As this region represents what are now the world’s largest producers of heroin post-ban, the impact of the Taliban’s decree is likely more muted than say Europe, where the majority of heroin is sourced from Afghanistan European Union Drugs Agency 2026. While countries like Myanmar and Laos are major suppliers of heroin to Australia, heroin from South-West Asia has historically accounted for approximately 20% of border seizures.⁵

Even if supply of the drug was not diminished greatly by the Taliban’s ban, the price impact would impact all purchasers of the drug, further supporting the idea that it is financial mechanisms and not lack of availability driving the effects shown in the paper.

Another unique feature of the Australian drug market is the drug of choice. Unlike Europe and the United States, where opiates are the illicit drug of choice, methamphetamine is the main illicit drug consumed. Historically, Australia had a large heroin population operating within the country. As documented in Moore and Schnepel 2021 several events transpired in 2001 to largely cutoff the nation from international supplies of heroin. This led to a rapid reduction in use across the country. Despite the reduction there has remained a small but persistent population of heroin users across the country.

The majority of the Australian population is located around the capital cities of each state which are all on or near coastal areas. One of the important aspects of the wastewater data used in this paper is the identification of variation in the intensity of consumption and drugs used across locations. When analyzing wastewater data, geographic variation in the types of drug consumed stands out.

⁵Australian Criminal Intelligence Commission 2021

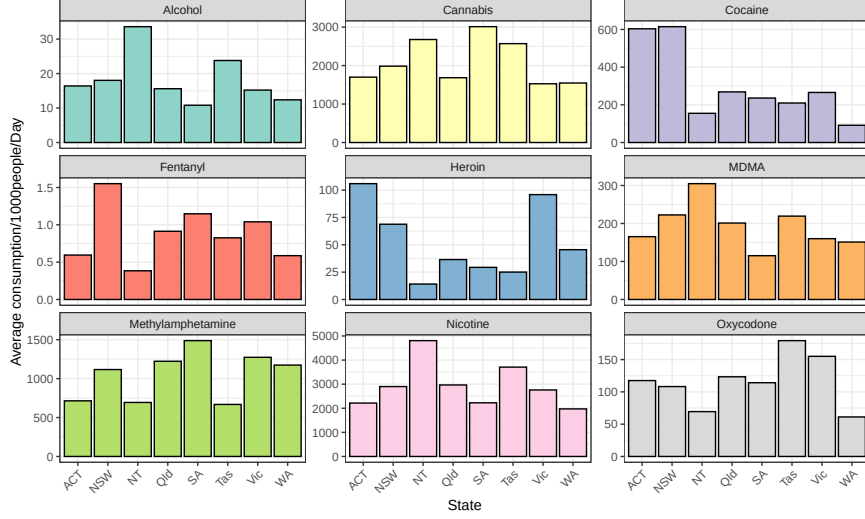


Figure 2: Average daily consumption by drug: Australian States and Territories

Notes: Figure 2 shows average consumption levels across Australian states and territories by drug type. Alcohol is reported in litres/1000people/day with all other drugs reported as mg/1000people/day. Due to differences in dosage, scales should not be compared across drug types.

Figure 2 shows average daily consumption of drugs across all Australian states and territories for 9 drug types as measured from the wastewater monitoring program. While heroin has little use among the population in Northern Territory, they have the largest consumption levels for alcohol, nicotine and MDMA on a per-capita basis. The Australian Capital Territory (ACT) and Victoria both stand out as having the highest per-capita heroin consumption. This geographic variation in the level of per-capita consumption is leveraged to amplify the fact that different regions may not react in the same way to an identical shock.

3 Tipping model for crime

The motivation for this paper is guided by the work of Goldstein [1985](#) who defines a three part framework to explain the causes of violence in drug markets. The first reason is psycho-pharmacological where it is the result of short of long term ingestion of intoxicating substances that induce violence or crime. The second is economically compulsive crime where some drug users engage in economically motivated crime in order to fund their use. The final reason is systemic, where crime or violence may occur as a result of the culture and standards users and sellers operate in. The purpose of this research is to focus on the second of this three-part framework, economically compulsive crimes.

Consider an individual who consumes a harmful good (e.g., an addictive substance) on a regular basis. The good is purchased out of available resources, which can be obtained either legally or illegally. When the price of the good increases, legal income alone may no longer suffice to finance the desired level of consumption, and the individual may then turn to illegal activities to supplement their budget.

Formally, consider an agent with an initial stock of resources B and an addiction level $s \in [0, 1]$. The agent chooses between a legal and an illegal source of income. If the agent works legally, they earn a wage

w . If the agent engages in illegal activity, they obtain an illegal payoff b , but face a probability of capture $r \in [0, 1]$. If captured, they incur a punishment cost C , which can represent fines, imprisonment, loss of reputation and so forth.

The agent's utility from legal activity is given by

$$U_{\text{legal}} = V(w) - ps,$$

Where p is the price of the harmful good, and ps represents the, addiction-scaled, expenditure or disutility associated with financing consumption through legal means.

If the agent chooses the illegal option, expected utility is

$$U_{\text{illegal}} = (1 - r)[V(b) - c(s) - ps] + r[V(0) - C],$$

Where $c(s)$ is the, potentially addiction-dependent, non-pecuniary cost of committing the crime (e.g., moral cost).

The agent commits the crime if

$$U_{\text{illegal}} > U_{\text{legal}},$$

or explicitly,

$$(1 - r)[V(b) - c(s) - ps] + r[V(0) - C] > V(w) - ps.$$

Rearranging, the condition can be written as

$$rps > V(w) - (1 - r)[V(b) - c(s)] - r[V(0) - C].$$

Thus, the tipping price p^* above which the agent is willing to engage in crime is

$$p > p^*(s, r, C) = \frac{V(w) - (1 - r)[V(b) - c(s)] - r[V(0) - C]}{rs}. \quad (1)$$

Assuming that the legal and illegal payoffs (w, b) and the non-pecuniary crime $c(s)$ are held fixed, (1) implies that as $(s \rightarrow 1)$, the threshold price p^* at which the agent turns to crime generally falls. Individuals with higher addiction levels thus become willing to commit crime at lower price levels of the harmful good. This effect is reinforced if $c(s)$ is decreasing in s : higher addiction lowers the intrinsic disutility of crime, further reducing the tipping price. Similarly, for a fixed level of s increases in the market price will tip the marginal person into engaging in crime.

Now allow punishment levels and probabilities of capture to vary by crime type $j \in J$. Suppose each crime type j is characterized by an illegal payoff b_j , a capture probability r_j , and a punishment cost C_j . Then the tipping price for crime type j is

$$p_j^*(s, r_j, C_j) = \frac{V(w) - (1 - r_j)[V(b_j) - c(s)] - r_j[V(0) - C_j]}{r_j s}. \quad (2)$$

This expression makes clear that, for an individual on the margin of offending, crimes with lower expected sanctions—i.e., lower (r_j) and/or lower (C_j) , holding $(w, b_j, c(s))$ fixed—tend to have a lower tipping price p_j^* . Consequently, as the price of the harmful good rises and individuals become more likely to consider illegal means of financing consumption, they will optimally select crime types with relatively low effective expected punishment. In equilibrium, marginal offenders should therefore be disproportionately observed committing offenses with lower probabilities of capture or lower punishment severity.

Where,

$$p_j^*(s, r_j, C_j) = \underbrace{\frac{V(w) - V(b_j) + c(s)}{r_j s}}_{\text{depends on } r_j} + \underbrace{\frac{V(b_j) - c(s) - V(0) + C_j}{s}}_{\text{depends on addiction level } s}.$$

This makes the comparative statics transparent. First, the effect of the capture probability:

$$\frac{\partial p_j^*}{\partial r_j} = - \frac{V(w) - V(b_j) + c(s)}{r_j^2 s}.$$

Under the assumption that

$$V(b_j) - c(s) > V(w),$$

we have

$$\frac{\partial p_j^*}{\partial r_j} > 0.$$

Thus, a higher probability of capture r_j raises the tipping price p_j^* ; equivalently, crimes with a lower probability of capture have a higher tipping price and remain attractive up to higher prices of the harmful good.

Next, consider the effect of punishment severity. From the expanded form,

$$p_j^*(s, r_j, C_j) = \frac{V(w) - V(b_j) + c(s)}{r_j s} + \frac{V(b_j) - c(s) - V(0)}{s} + \frac{C_j}{s},$$

so

$$\frac{\partial p_j^*}{\partial C_j} = \frac{1}{s} > 0.$$

Therefore, a higher punishment cost C_j raises the tipping price, whereas crimes with lower punishment severity C_j have lower tipping prices and are decreases in s . This shows that the severity of punishment may serve as less of a deterrent for individuals with higher levels of addiction and allows the prediction that given an increasing price, we should expect individuals on the margin of committing a crime to be pushed to crimes with a lower expected punishment and probability of capture such as theft and other petty crimes.

4 Data

I rely on data from three sources. Price data from the UNODC, wastewater data from a national program in Australia, and crime data collected by Australian state police forces.

4.1 Drug Prices

The UN’s Office of Drug Crime (UNODC) provides worldwide monitoring of drug prices throughout the world. To be included in the dataset a participating country’s law enforcement agency submits prices they have identified in their own countries either through drug seizures or undercover purchases of drugs. While the data is often sparse, in some cases, such as in Afghanistan, the UNODC will conduct special projects to monitor areas deemed problematic. The UNODC has collected monthly farm-gate prices for opium and market prices for heroin at the wholesale level (\$/kg) and estimates for crop cultivation through a combination of satellite imaging and on the ground reporting to track changes in the heroin and opium market within the country.

Details on prices in illicit drug markets are notoriously difficult to obtain. The data from the UNODC primarily reflects changes in the price of wholesale drugs. While some retail prices are provided, they are either lagged prior to the ban or not provided with a sufficient enough time span for this analysis. As such, the analysis will rely primarily on price changes in the wholesale market.

4.2 National Wastewater Drug Surveillance Program

Wastewater monitoring initiatives in Australia began in 2009. In response to perceived escalating drug issues within the country, the Australian government implemented the National Wastewater Drug Surveillance Program in 2016⁶. This program is notable for being the most comprehensive globally in terms of the variety of drugs measured, length of operation and the frequency of testing. The Australian Criminal Intelligence Commission (ACIC) administers the program, with analytical support from several Australian universities⁷, ensuring coverage in all Australian states.

Approximately 50 wastewater treatment plants (WWTPs) participate in the program, representing around 50-60% of the Australian population. Each wastewater testing area corresponds to a specific sewage catchment zone. Although these catchment areas do not align strictly with defined neighborhood boundaries, they effectively delineate sections of urban environments⁸. WWTPs may periodically enter or exit the program, resulting in minor fluctuations in the population under surveillance. Wastewater sampling is conducted in every state capital, along with several regional locations⁹. While the exact locations of the WWTPs are confidential, sites are coded as either capital city or regional location and their corresponding states are provided. Since each state contains one capital city, this facilitates the identification of capital city locations. Regional testing sites, being unidentifiable, are omitted from the analysis, ultimately including around 20 WWTPs in the study.

A crucial point is that while approximately half of the Australian population is monitored, nearly 100% of individuals residing within the designated geographic areas are included. For individuals within a sewage

⁶Report 1 of the National Wastewater Drug Monitoring Program **ACIC1**

⁷University of Queensland and University of South Australia

⁸Visual example of a catchment zone is listed in Appendix

⁹Regional locations are defined as non-capital cities, and are typically large population centers within each state.

catchment area, inclusion in the sample hinges solely on their use facilities that feed into the wastewater system. Moreover, while regional locations are excluded from the crime analysis, they represent a small fraction of the overall sample population. According to the 2021 Australian Census, the cumulative population of the corresponding capital cities is approximately 13.3 million. Thus, although the program covers about half of the Australian population, around 85% of capital city residents are represented in the sample ¹⁰.

In order to produce valid estimates for consumption, daily composite samples are collected by participating WWTPs over seven consecutive days. Composite sampling over consecutive days is important to account for variation consumption patterns based on the time of day or day of the week¹¹. The composite samples are then sent to a lab to analysis where specific metabolites can be identified and their proportions to the overall sample calculated. Given the metabolite type, proportion of metabolite in the sample, and the population servicing each catchment area, scientists can create estimates for total amount of a specific drug consumed in a given time period. In certain instances, collecting samples for a full seven days may be impractical due to weather conditions or special events that could distort results. The primary objective of wastewater monitoring is to provide a reliable estimate of regular drug use within a specific area. For example, an event such as a music festival within a city’s catchment area could result in an influx of users of the municipal sewage system and an uptick in drug consumption, potentially skewing conclusions regarding drug use in the broader population. Additional details regarding the sampling protocol and relevant quality controls are discussed in Irvine et al. (2011)Irvine et al. 2011, Lai et al. (2011)Lai et al. 2011, Lai et al. (2015), Tscharke et al. (2016)B. J. Tscharke et al. 2016, and Bade et al. (2019)Bade et al. 2019.

¹⁰see figure 5 in appendix A for population coverage details

¹¹See Appendix for examples of intra-week consumption variation

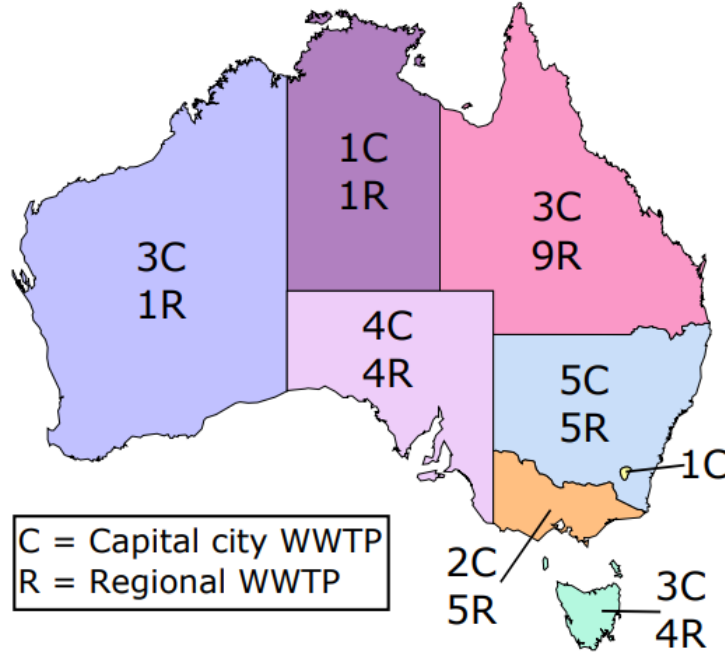


Figure 3: Testing site coverage

Notes: The figure shows the geographic coverage of the wastewater monitoring program in Australia^a. States and territories are divided by color and list the number of testing sites operating in each region. C indicates a site in a capital city and R a regional location. Regional is defined as non-capital city. Exact locations for each site is not known, but because capital cities are unique to each state data from these locations can be linked to demographic and crime data at the capital city level.

^aGraph presented in report 21 of the National Wastewater Drug Monitoring Program

The samples collected are normalized by population to estimate drug consumption per 1,000 individuals per day, reported tri-annually for capital cities and regional locations. As testing sites provide repeated sampling over time, trends within specific states will accurately reflect changes in consumption within the corresponding catchment areas over time. Given that the catchment areas remain constant in the study, fixed effects can be applied to account for characteristics unique to each area.

4.3 Crime Data

Data on reported crimes are collected from Australian State police jurisdictions. Uniform crime reporting data is not available in Australia at the monthly level which requires monthly reporting of incidents to be collected from each state and territory police agency. All states and territories in Australia with the exception of Tasmania are included¹². The data reported by state police record a monthly count of reported crimes by crime type. While each state and territory follows similar national reporting guidelines, the granularity of crime reported differs. This results in different levels of aggregation across geography. In order to make levels comparable across geography, crimes are aggregated to general categories reported across all jurisdictions.

¹²Tasmania is a special case and a rural, isolated population. They are unlikely to be impacted heavily by the effects of the Taliban's ban and are not main drivers of any crime trends within the country. Yearly trends are shown in Appendix X to confirm no major changes in Tasmanian crime trends following the ban's announcement.

Crimes are first classified as “financially-motivated” and “personally-motivated”. The determination for classification of “financially-motivated” crimes is if money is the main result of the crime. Under this criteria crimes such as burglary, robbery, theft, larceny etc.,. Similarly “personally-motivated” crimes are typically described as causing physical harm without financial reward. Crimes such as homicide, assault, and sexual crimes meet this criteria.

Since the effects of a price increase may impact crimes differently, crimes are further broken down into sub-types. Personally motivated crimes are separated into “Assault” and “Homicide”, while financially motivated crimes are separated into “Theft”, “Burglary”, and “Robbery”. Crimes have differences in both moral and societal stigma as well as punitive outcomes. While both are considered societal taboos, it is unlikely that a person committing theft, a non-violent crime, is held to the same standards as someone committing a robbery, which while it may be non-violent often uses the implication of violence to succeed.

Crimes are reported as counts. Counts are normalized to local populations to report the rate of crime per month per 1,000 people. Additional demographic and socioeconomic variables are collected from the Australian Bureau of Statistics (ABS) to account for changes in age profiles, employment levels, gender composition, and marriage rates¹³.

5 Estimation

Estimation follows in two parts to identify two separate research questions. The first is identifying the impacts the increase in global heroin prices has on local consumption of drugs in Australia. In this estimation I aim to identify own price elasticity for heroin as well as cross price elasticities for other drugs to suggest substitution patterns. Becker, Murphy, and Grossman 2006 motivate this well by identifying conditions for consumer elasticity necessary for enforcement actions to be impactful. In the second estimation, I use the finding from the price elasticity estimation to motivate analysis on local crime rates. Given low price elasticities, increased prices should induce higher financially motivated crime at the local level, supporting the economic compulsive mechanisms between drug use and crime as described in Goldstein 1985.

A unique aspect of the wastewater data is that it highlights geographic variation in consumption across both the intensity of consumption and type of drug consumed. Figure 4 presents average consumption of heroin across 8 capital cities in Australia. Since sampling is already population normalized to mg/1000people/day the results in the figure present relative intensity of consumption in communities across Australia. What immediately becomes apparent is the variation in consumption. Melbourne has by far the largest level of consumption at approximately XXXX mgs/1000people/day where Darwin, Northern Territory has essentially no consumption.

¹³specific variables are the percentage of the population in specific 5-year age buckets age 15-65. The proportion of the population that is male, the proportion married, and the proportions of the population fully employed, unemployed, and not in the labor force.

5.1 Price Elasticities & Substitution

I am interested in identifying the demand responses for a variety of drugs in response to the price increase in global heroin prices following the announcement of the Taliban’s ban. An ideal experiment to use would be an equation of the form:

$$\ln(Q_{wt}^h) = \beta_{OLS}^h \ln(P_{w,t-j}^h) + \gamma X_{st} + \epsilon_{st} \quad (3)$$

Where Q is consumption of heroin as reported at wastewater testing site w , $P_{s,t-j}^h$ is the j -month lagged retail price of a pure dose of drug (i), and ϵ is the error term. Typically, researchers rely on purity adjusted prices to take into account the stickiness of nominal prices in the retail market. It is common for nominal prices a retail customer faces to remain constant over time. What does fluctuate is the purity of the product they are offered, impacting the implied pure-weight price of a retail drug over time. Evidence from samples of police seizures, as well as survey evidence from people who inject drugs support this retail price stickiness, detailed retail prices for heroin are unavailable in Australia. As such, I rely on prices of opium and heroin in Afghanistan as a proxy for the presumed purity adjusted price increases in Australia.

As the international prices of heroin and opium increase, financial pressures on suppliers would induce lowering of purity at the retail level. I use the monthly price of opium in this estimation to instrument for the degree of financial pressure facing suppliers as they are forced to either lower purity or trim their margins in order to maintain the same level of profitability. The sustained nature of the ban helps to support the fact that suppliers must make adjustments to the products offered and cannot wait out the ban.

I rely on the sudden supply shock following the Taliban’s announcement of the ban in April of 2022. Due to the international aspect of the ban’s impact, it is assumed all locations in Australia are impacted by this ban leaving no control sample. To overcome this issue, I employ a continuous difference-in-differences estimation method by utilizing the geographic variation in pre-ban heroin consumption levels across capital cities in Australia.

$$\ln(Q_{wt}^i) = \beta_{IV}^i \ln(P_{s,t-j}^o) * Q_w^H + \gamma^i X_{st} + \delta_w + \tau_m + \epsilon_{wt}^i \quad (4)$$

Figure 4 highlights the differences in heroin consumption prior to the ban. As all measures are already population normalized, these levels can be thought of as a population level dosage of heroin consumption. Q_w^H is a time-invariant measure of pre-shock heroin usage as recorded in the wastewater data. Location fixed effects (δ_w) absorb all time invariant differences across catchment areas, essentially serving as neighborhood fixed effects, while month fixed effects (τ_m) account for seasonality in both the production and consumption of drugs. β then measures the differential price elasticities w.r.t opium prices for higher and lower consumption areas. The vector X is comprised of state level controls including the log of the local unemployment rate, logged not in labor force (NILF) rate, proportions of the population married and the proportion of the population that is male.

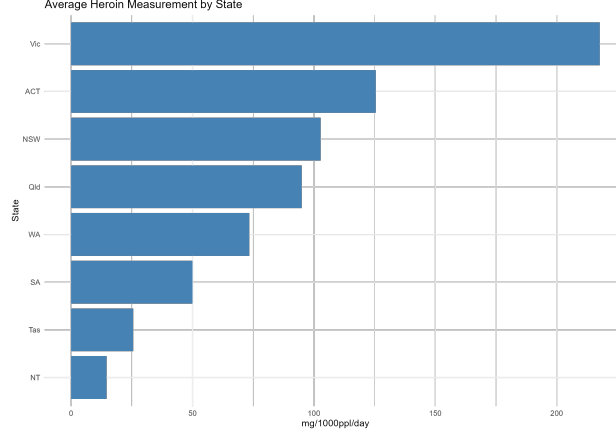


Figure 4: Average daily heroin consumption: Capital Cities

Notes: Figure 4 shows average consumption levels of heroin across all eight capital cities. Consumption is reported as mg/1000people/day.

To ease interpretation of the coefficient on the interaction, I transform the heroin consumption variable from daily milligrams consumed per 1000 people to daily doses consumed per 1000 people ¹⁴. The coefficient β can now be interpreted as the change in local crime for a change in the log price *and* a one dose increase in daily heroin consumption pre-ban. This serves as an appropriate proxy for the intensity of addiction in capital cities across Australia.

5.2 Crime Estimation

Due to the global impact of the Taliban's ban, all heroin users within Australia are treated. To separate the impact of price increases on the propensity for crime, I employ a similar estimation strategy to equation (4). Using the level of consumption in each Australian capital city prior to the ban creates a dosage framework for treatment. Some cities, like Sydney have nearly triple the concentration of heroin in the wastewater than cities like Darwin, NT or Adelaide, SA. Because the wastewater is already population normalized this can directly be translated into a per-capita level of heroin exposure in each city. I use the following form to estimate the effect of increased exposure to the heroin price increases, due to pre-ban heroin demand, on the local crime rate for crime j .

$$\ln(Crime_{ct}^j) = \beta_{CDD}^{i,o} \ln(P_{s,t-j}^o) * Q_c^h + \gamma^i X_{st} + \delta_w + \tau_m + \epsilon_{wt}^i \quad (5)$$

Where Q_c^h represents the pre-ban level of heroin in city c . I first run specifications for crimes broadly broken into personally and financially motivated crimes. These categories are determined by whether the outcome is deemed to have a financial or personal motivation as its outcome. Crimes such as theft, robbery, burglary, and prostitution would meet the definition for financially motivated crimes. Crimes such as assault, homicide, stalking and rape are deemed personally motivated.

¹⁴Metabolite specific details and dosages are detailed in Appendix X

I further separate crimes into subcategory to understand if specific crime types are driving trends. Indeed, allowing for heterogeneous probability of capture and/or punishment in equation (1) supports this as the tipping price increases with severity of punishment or probability of capture. I narrow the analysis to five main groups ; “Assault”, “Homicide”, “Burglary”, “Robbery”, “Theft”. It would be reasonable to assume differential effects across these types. For instance, while assault and homicide are both considered personally motivated crimes, the punishment, investigative effort from police, and social taboo are much higher for homicide than assault suggesting a higher tipping price and thus a more muted impact of price increases on its probability of occurring.

A concern in estimation is the proximity of the Taliban’s announcement of the ban to the COVID-19 pandemic. The main reason for this was how Australia handled the pandemic. The Australian government imposed a short-term national lockdown early on, but beyond this national lockdown, there were state specific lockdowns based on local COVID case numbers. This resulted in different states being in lockdowns at different times and for varying lengths ¹⁵. As local lockdowns are likely to have an impact on local crime rates, the crime data is filtered to only include dates after 2021 and until 2025.

6 Results

I examine how the sudden increase in opium prices stemming from the Taliban’s ban impacted drug markets and crime rates in Australia. The motivations for these are guided by several predictions. The first is that heroin is an inelastic drug. Due to the physical addictive-ness of the drug, users of the drug are likely less influenced by price movements and rather the prospects of facing withdrawal symptoms when weighing their consumption decisions. The second is that if substitutes are available, some users will switch consumption to other, possibly more harmful drugs. The third is that, given a low price elasticity, a user will be faced with an increased financial burden to maintain a given level of consumption. This increased financial burden may induce users to commit crimes in order to supplement their budgets.

6.1 Price elasticities

XXXX Preliminary Results XXXX

I begin by examining the impacts of the price increase on heroin consumption within Australia. Table 1 summarizes the effects across three specifications. Columns (1) and (2) show the raw effect of the global price of opium on consumption of heroin in Australia. The point estimates are statistically insignificant but both centered around zero. Columns (3-5) add the continuous treatment interaction of the price of opium over time with the pre-shock level of heroin consumption in each catchment area. Column (3) is run without any fixed effect. The coefficient on price is now significant at -0.1652 with no significance on the interaction term. Column (4) is run without controls but including month and catchment area fixed

¹⁵See figure X in Appendix X for details of the national and state lockdowns

effects. This produces significant coefficients on both the lagged price trend and the interaction term. The interaction term has a positive coefficient indicating lower price elasticities in catchment areas with higher pre-shock heroin use. The last specification, column (5), includes both controls and fixed effects. For this specification, price produces a coefficient of -0.125, although it is no longer significant. The interaction term remains significant at a coefficient of 0.0009.

Table 1: Heroin Consumption

	Log avg. consumption Heroin				
	(1)	(2)	(3)	(4)	(5)
Log opium price (t-3)	-0.0704 (0.0500)	-0.0199 (0.1261)	-0.1652** (0.0820)	-0.1410** (0.0653)	-0.1251 (0.1350)
Log unemployed		0.4271** (0.2116)			0.3369 (0.2136)
Log NILF		1.623 (1.302)			2.380* (1.372)
Log pop. 15–49		0.5666 (3.050)			0.6507 (3.115)
Constant			4.213*** (0.4084)		
Pre-shock heroin			0.0439 (0.0655)		
Log opium (t-3) $\times$ Pre			0.0178 (0.0132)	0.0176* (0.0104)	0.0189* (0.0104)
Month FEs	Yes	Yes	No	Yes	Yes
Catchment area FEs	Yes	Yes	No	Yes	Yes
Observations	327	327	318	318	318
R ²	0.62770	0.63756	0.34585	0.61676	0.62796
Within R ²	0.00652	0.03284		0.01613	0.04489

*Notes: Estimates report the effect of opium ban on heroin consumption, measured by monthly wastewater concentrations. Consumption is reported in mg/1000ppl/day. Column 1 has no FE or controls, column 2 includes month FE and catchment area FE to control for seasonality in drug consumption and production and location specific effects. Standard errors are clustered at the catchment area level. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.*

These results are generally lower than those reported summarized in Gallet 2014. It is difficult to tell from the estimation alone how much of the reduction is driven by changes to the price and what is driven by possible reductions in the overall availability of heroin to retail consumers.

In the 2001 Australian heroin drought Moore and Schnepel 2021 identify lack of availability as a main driver of the results, using national surveys to show that 88% of respondents reported heroin was difficult to obtain, in contrast to previous years survey where 99% of respondents reported the drug was easy to obtain. Using the same survey, no dramatic change in availability was reported during this period, suggesting that it is changes in the wholesale prices, and not changes local availability driving any changes in consumption.

XXXX Check Survey Results XXX

A large question in the literature is how interconnected drug markets are and to what degree users switch between drugs. Table 2 presents the effects of the opium shock on the licit drug market. Column (1) measures the effects on nicotine use and column (2) shows the effect on alcohol use. Measurements in the wastewater differ slightly between these two. As with all other drugs shown, nicotine is reported in *mg/1000people/day* while alcohol is measures in *litres/1000people/day*. Labor market and demographic controls are included in both specifications in addition to month and catchment area fixed effects.

Table 2: Licit Consumption

	Log avg. consumption	
	Nicotine	Alcohol
	(1)	(2)
Log opium price (t-3)	-0.0686* (0.0403)	-0.1524*** (0.0438)
Log unemployed	0.0502 (0.0639)	-0.0297 (0.0696)
Log NILF	-0.9481** (0.4094)	-1.740*** (0.4463)
Log pop. 15-49	1.889** (0.9269)	1.144 (1.014)
Log opium (t-3) $\times$ Pre	0.0088*** (0.0031)	0.0110*** (0.0033)
Month FEs	Yes	Yes
Catchment area FEs	Yes	Yes
Observations	316	316
R ²	0.71428	0.76060
Within R ²	0.05773	0.15591

*Notes: Estimates report the effect of opium ban on licit drug consumption, measured by monthly wastewater concentrations. Column 1 measures nicotine consumption in mg/1000ppl/day. Column 2 measures consumption in litres/1000ppl/day. Both specifications include month and catchment area FEs. Standard errors are clustered at the catchment area level. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.*

In both specifications the coefficients on price are negative and significant. The size of the magnitudes vary, with a coefficient on -0.0686 on nicotine and -0.1524 on alcohol. The interaction terms are positive and significant in both specifications as well. While the size of the effect is smaller than the specification in Table 1, the direction is the same. If nicotine and alcohol are complements to heroin, as the price trends would indicate, the mirrored directionality on the interaction terms helps to bolster the idea that areas with higher pre-shock levels of addiction exhibit lower price elasticities.

Table 3: Illicit Consumption

	Log avg. consumption					
	Cannabis (1)	Cocaine (2)	Fentanyl (3)	MDMA (4)	Methamphetamine (5)	Oxycodone (6)
Log opium price (t-3)	0.0470 (0.0526)	-0.2412*** (0.0873)	-0.0660 (0.0973)	0.2263 (0.1372)	0.2983*** (0.0649)	-0.1384** (0.0539)
Log unemployed	-0.2552*** (0.0834)	0.3017** (0.1381)	0.3276** (0.1539)	1.380*** (0.2172)	0.1449 (0.1026)	0.1856** (0.0852)
Log NILF	1.589*** (0.5339)	-1.368 (0.8872)	-1.687* (0.9886)	-1.396 (1.395)	-0.8138 (0.6591)	-3.118*** (0.5476)
Log pop. 15-49	-1.538 (1.213)	11.61*** (2.014)	0.3029 (2.244)	-6.179* (3.167)	-4.922*** (1.496)	0.3749 (1.243)
Log opium (t-3) $\times$ Pre	0.0109*** (0.0040)	0.0055 (0.0067)	0.0146* (0.0075)	0.0101 (0.0105)	-0.0014 (0.0050)	0.0111*** (0.0041)
Month FEs	Yes	Yes	Yes	Yes	Yes	Yes
Catchment area FEs	Yes	Yes	Yes	Yes	Yes	Yes
Observations	316	318	318	317	318	318
R ²	0.84027	0.70583	0.63208	0.45117	0.66854	0.76937
Within R ²	0.14569	0.16127	0.05092	0.18266	0.09444	0.25954

Notes: Estimates report the effect of opium ban on illicit drug consumption, measured by monthly wastewater concentrations. All columns report consumption in mg/1000ppl/day. Both specifications include month and catchment area FEs. Standard errors are clustered at the catchment area level. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

Table 3 presents results for illicit drugs. The same specification in Table 2 was run on a variety of illicit drugs. Column (1-5) present results for cannabis, cocaine, fentanyl, MDMA, methamphetamine, and oxycodone. Fentanyl and oxycodone are both opioids in the same drug class as heroin. While the Australian heroin market is smaller than that of Europe or the United States, one of the features that defined the American opioid crisis was substitution between these drugs. Indeed, the introduction of fentanyl into U.S. illicit opioid markets increased the overdose rate dramatically. While we see decreases in opioid consumption overall for these two categories, the interaction of price and pre-shock exposure produces positive and significant results for both drugs. This presents a worrying trend with indications of switching to the more harmful fentanyl supply possible mimicking trends seen in the U.S. opioid market.

Given this specification, it is not possible to fully separate out what trends are driven by supplier decisions and what are purely changes to demand. It is highly likely that, given changes to price and availability in the wholesale market, retail suppliers have an increased incentive to switch to the cheaper and easier to transport fentanyl.

Column (1) also presents positive results for the interaction on cannabis. Substitution from opioids into cannabis has been documented and has been suggested as a way for opioid users to reduce consumption.

6.2 Crime

XXXX Preliminary Results XXXX

I next move to estimating the effects of the price increase on local crime rates. The results from table 1 support the fact that demand for heroin is inelastic. Given an inelastic good, price increases raise the financial strain on consumers. Given fixed budgets, physically addicted consumers would need to find ways to supplement their incomes and, given the likely legal labor market constraints users of heroin face, user will switch to the illegal market. Table 4. presents the effects of the price-shock on crime rates in capital cities across Australia.

Due to the fact that the exact locations of the catchment areas used for consumption measures are not known to me, I aggregate pre-shock heroin consumption to the capital city level. I run this specification for seven Australian capital cities, with Hobart, Tasmania left out due to lack of data. Crimes are grouped by broad crime type into financially motivated and personally motivated crimes. The classification of these crime types is determined by the ultimate result of the crime. Financially motivated are defined as resulting in monetary profit for the perpetrator and personally motivated is a crime on a person that does not results in monetary profit for the perpetrator. Crimes are normalized to the city population and reported as *crime/1000people/month*. Column (1) presents the results for financially motivated crimes. Similar to the coefficients in the elasticity specifications, the coefficients on price and the interaction term are significant with opposite signs. Price has a coefficient of -0.0775 indicating a general decrease in crime with respect to price with the positive interaction term indicating a muted effect in areas with higher pre-shock consumption of heroin.

Table 4: Financially vs. Personally Motivated Crimes

	Log financial crime rate Financially Motivated (1)	Log persons crime rate Personanlly Motivated (2)
Log opium price (t-3)	-0.0766** (0.0299)	-0.0301 (0.0228)
Log unemployed	-0.1596*** (0.0445)	-0.0451 (0.0339)
Log NILF	-0.0899 (0.2667)	-0.7608*** (0.2033)
log(TotalPop)	-13.77*** (3.868)	3.423 (2.949)
ProportionMarried	0.8348 (0.8068)	2.111*** (0.6152)
ProportionMale	20.69*** (7.564)	-0.9366 (5.768)
Log pop. 15-49	11.60*** (3.024)	0.5772 (2.306)
Pre-shock heroin $\times$ Log opium price (t-3)	0.0164*** (0.0028)	-0.0050** (0.0022)
Month FEs	Yes	Yes
State FEs	Yes	Yes
Observations	322	322
R ²	0.97661	0.98488
Within R ²	0.16715	0.37712

*Notes: Estimates report the effect of opium ban on prevalence of crime reported as crime/1000ppl/month. Data is limited to 7 Australian capital cities (excluding Hobart, Tasmania) Financially motivated includes all crimes with profit as the main outcome of the crime. Personally motivated includes crimes on a person, not included in Financially motivated. Both specifications include month and catchment area FEs. Standard errors are clustered at the city area level. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.*

Column (2) reports the effects on personally motivated crimes and primarily serves as a robustness check. If the mechanism for varying crime rates based on pre-shock heroin consumption is financially focused then we should not see increases in personally motivated crimes. Indeed the results show a negative, but statistically insignificant effect on price, but a negative and statistically significant coefficient on the interaction term.

The results in table 2 support the theory that financial burdens can induce crime as users seek to supplement their incomes through illegal means. We should not expect uniform effects across crime types however since the consequences, both legally and morally, vary by crime type.

Table 5: Crimes by Type

	Assault (1)	Burglary (2)	Log_Count Homicide (3)	Robbery (4)	Theft (5)
Log opium price (t-3)	-0.0280 (0.0678)	-0.1031 (0.0694)	-0.0640 (0.0982)	0.0032 (0.0820)	-0.0802 (0.0567)
Log unemployed	-0.0526* (0.0248)	-0.0893 (0.1159)	0.1880 (0.2373)	0.0493 (0.1663)	-0.1082* (0.0523)
Log NILF	-0.7446* (0.3386)	0.2644 (0.6389)	0.1193 (1.114)	-1.093 (0.7421)	-0.6842 (0.5523)
log(TotalPop)	4.081 (10.60)	5.879 (5.318)	-17.54 (10.98)	-1.666 (5.819)	-9.478 (8.104)
ProportionMale	-0.3958 (15.32)	28.51** (10.18)	46.18 (27.55)	54.85*** (5.037)	18.46 (16.70)
ProportionMarried	1.648* (0.7407)	-0.0155 (1.162)	-4.296 (3.035)	0.1842 (2.292)	0.0386 (0.9221)
Log pop. 15-49	0.5966 (7.807)	-6.275 (4.980)	12.01 (9.630)	1.639 (5.178)	10.09 (6.132)
Pre-shock heroin $\times$ Log opium price (t-3)	-0.0047 (0.0069)	0.0275*** (0.0047)	0.0200** (0.0080)	0.0126*** (0.0033)	0.0145* (0.0072)
Month FEs	Yes	Yes	Yes	Yes	Yes
State FEs	Yes	Yes	Yes	Yes	Yes
Observations	335	335	275	335	335
R ²	0.99444	0.98544	0.77237	0.94728	0.99272
Within R ²	0.49464	0.13159	0.02644	0.10116	0.38941

*Notes: Estimates report the effect of opium ban on prevalence of crime reported as crime/1000ppl/month . Data is limited to 7 Australian capital cities (excluding Hobart, Tasmania). Columns report crime rates by subtype. Differences in police reporting require aggregation in some cities. Standard errors are clustered at the city area level. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.*

Table 5 presents the same specification with results separating into subtypes. Columns (2,4,5) all report the results for financially motivated crimes. None of these specifications show significant results on the broad impact of opium prices on crimes of these types. All of these show positive and significant effects on the interaction. For a 1 dose increase in heroin consumption per 1000 people, burglary increases 0.0275, robbery by 0.0126, and theft by 0.0146.

Surprisingly homicide also has a positive effect of 0.02. It is possible changes in the homicide rate are motivated by changes in the overall drug network within the country. Indeed, given increases in retail prices, drug turf may become more valuable leading to violence between sellers in the market. While descriptions of the homicides are not provided e.g. vehicular, gun violence, intimate partner, increased research into this effect is worthwhile.

7 Discussion & Conclusion

I provide new evidence of changes in consumption patterns for drugs following a large market shock. I also document apparent spillovers from this shock into the persistence of crime in local areas. The shock used in this study is ideal as it mimics an extreme form of punitive supply reduction policies that seek to reduce access of drugs to the public. Several key findings stand out.

Looking first at drug consumption, following the increase in prices consumption for heroin does reduce. Despite this reduction, price elasticity for heroin seems to be lower than most other research suggests. I also document that price elasticities appear to be lower in areas with higher pre-shock consumption highlighting that the level of addiction may influence responsiveness to market interventions. There appears to be fairly large rates of substitution into methamphetamine following the shock. Of special importance is substitution into other opioid drugs. Most notable is substitution into fentanyl which has a higher probability of overdose than heroin or oxycodone.

There are many reasons to suspect sellers would shift to synthetic drugs in the wake of the Taliban's ban. Opium fields take time to develop and farmable land of this crop requires either explicit government permission or weak oversight. It would be hard for DTOs to quickly shift production into another country. Given the significant geographic shift in demand for raw opium, Myanmar is only able to increase production by 35%. Synthetic drugs, such as fentanyl, are relatively easier to produce and transport.

The strengths of wastewater monitoring programs can also be a weakness as drugs are uniquely identified by their metabolites. Fentanyl, which is in the drug class of nitazenes or synthetic opioids, is identified by the metabolite carafentanyl. There are many nitazenes and it is common for suppliers to introduce new ones into drug market, many of which are even more powerful than fentanyl. Other drugs in the nitazene drug class have different metabolites so would be unidentified in the wastewater data. Reports of overdoses in Australia caused by nitazenes other than fentanyl have grown since monitoring began in 2021. It is then likely that the degree of substitution from heroin to other synthetic opioids is under measured in this paper.

Moving to crime, the results support the idea that financial pressure is a channel for crime prevalence. I identify that cities with higher levels of heroin consumption, prior to the Taliban's ban, saw increased rates of financially motivated crime as the international price for opiates increased. In these same areas there also is a negative effect on personally motivated crimes. When separating crimes into subtype, burglary saw the largest impact of the ban, with significant results on robbery and theft. Surprisingly homicide also sees an increase. The mechanism for homicide may be different and shocks to supply markets may induce seller side fight for supply and territory.

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A Appendix A

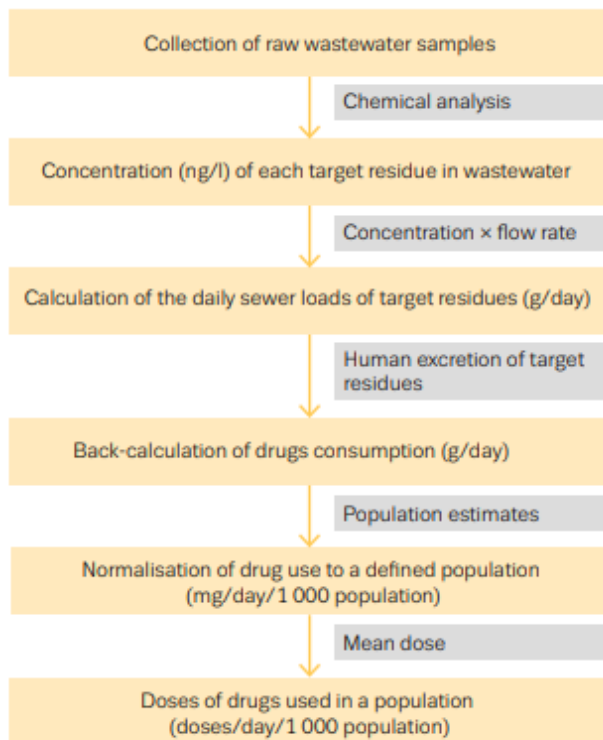
State or territory	April 2021 Capital	April 2021 Regional	June 2021 Capital
ACT	1	0	1
NSW	3	8	3
NT	1	1	1
Qld	3	8	3
SA	4	5	4
Tas	3	2	3
Vic	2	8	2
WA	3	4	3
Sites	20	36	20
Population (millions) C & R	11.2	1.8	11.2
% of Australian Population	47.9	7.9	47.9
Total population (millions)	13.0		11.2
% of Australian population	55.8		47.9

Estimates have been rounded to the nearest 0.1 million. Census 2016 population used (23,401,892) for population percentage estimates.

Figure 5: Population coverage: Wastewater Monitoring

FIGURE 1.1

The main consecutive steps of the wastewater analysis approach and the data required for each step



NB: Modified from Castiglioni et al., 2014.

Figure 6: Wastewater-based epidemiology workflow

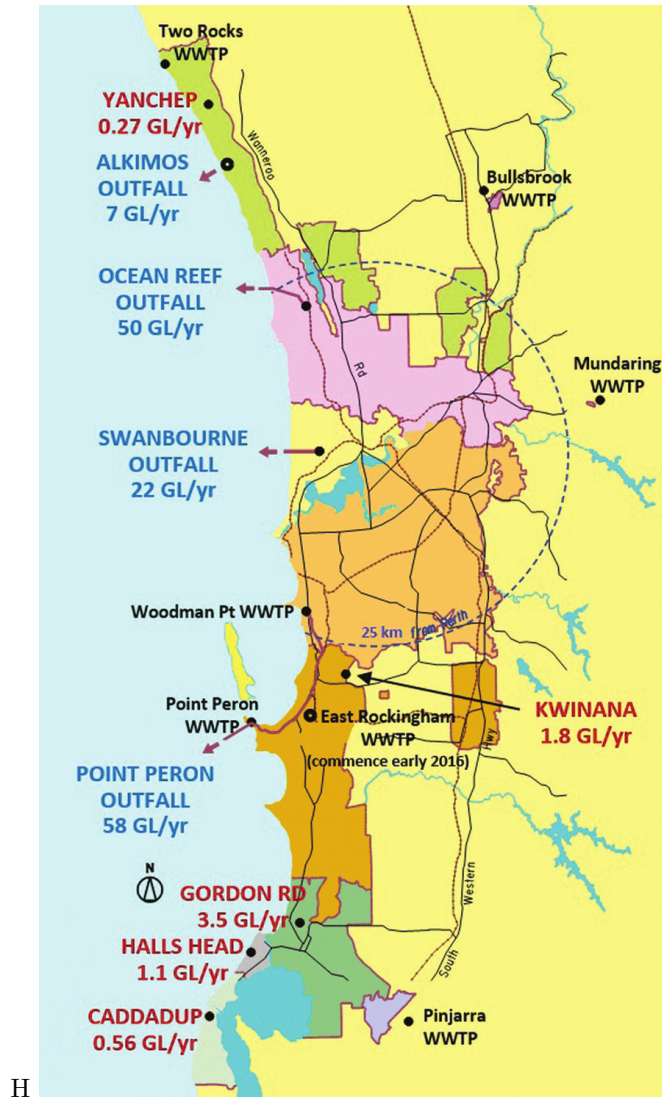


Figure 7: Perth, Western Australia catchment areas

Analyte/metabolite	Drug	Limit of detection (LOD) [ng/L]	Limit of quantification (LOQ) [ng/L]	Excretion factor	Standard dose pure drug (mg)
Amphetamine	Amphetamine	12	16	0.394 ^a	30 ^b
Cocaine	Cocaine	17	50	0.075 ^b	100 ^b
Cotinine	Nicotine	33	100	0.3 ^c	1.25 ^c
Norfentanyl	Fentanyl	0.1	0.1	0.3 ^d	0.2 ^d
MDA ^e	MDA	1	4	n.a.	n.a. ^e
MDMA	MDMA	1.5	2	0.225 ^b	100 ^b
Mephedrone	Mephedrone	0.4	0.8	n.a.	n.a.
Methylamphetamine	Methylamphetamine	33	100	0.39 ^d	30 ^b
Methylone	Methylone	0.01	0.1	n.a.	n.a.
Hydroxycotinine	Nicotine	17	50	0.44 ^c	1.25 ^c
Noroxycodone	Oxycodone	0.1	1	0.22 ^f	20 ^d
Ethyl Sulphate	Alcohol (ethanol)	167	500	0.00012 ^g	10g ^g
Benzoylcegonine	Cocaine	33	100	0.35 ^h	100 ^b
6-Monoacetylmorphine	Heroin	0.5	1.0	0.013 ^b	20 ^d
THC-COOH	THC (Cannabis)	30	180	0.1 ^h	8 ^h
Norketamine	Ketamine	1	2	n.a. ⁱ	n.a.

Figure 8: Metabolite details